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DSTALGO

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Data Structures and Algorithms

Data structure is basically how datas are being stored and organize inside a computer.

Algorithm is like the steps that the computer follows to solve something. Both of this is important because it helps the program work more better and easier.

Reference: https://opendsa-server.cs.vt.edu/ODSA/Books/wfu/csc201-b/fall-2026/201B/html/IntroDSA.html

Linear Data Structure

Linear data structure is when the datas are arrange one by one in a straight order. Some example of this are array, stack, queue and linked list.

Reference: https://www.geeksforgeeks.org/dsa/introduction-to-linear-data-structures/

One-Dimensional Array

One dimensional array is like a list where many values can be store in one line. Every value have its own index that can be use to access it.

Example:

[10, 20, 30, 40]

Reference: https://www.geeksforgeeks.org/dsa/types-of-arrays/

Big O Notation

Big O notation is use to know how fast or slow an algorithm can be when the datas gets bigger. It can also help compare which algorithm is more efficient.

Some examples is:

- *O(1) - constant*
- *O(n) - linear*
- *O(log n) - logarithmic*
- *O(n²) - quadratic*

Reference: https://cs50.harvard.edu/extension/2026/spring/notes/3/

Linear Search

Linear search is a searching method where it checks every value one by one until the value that you need is found. The data doesnt really need to be sorted for this.

Example:

[5, 10, 15, 20]

Find 15

5 = no

10 = no

15 = found

Reference: **https://opendsa-server.cs.vt.edu/ODSA/Books/scgssm/csc230/spring-2026/CS_Courses/html/BinarySearch.html**

Binary Search

Binary search is another searching method but the datas should be sorted first. It checks the middle value then it choose if it will search on the left or right side. It keeps doing this until the value is found.

Example:

[10, 20, 30, 40, 50]

Find 40

Middle is 30

40 is bigger than 30

Search right side40 is found

Reference: **https://opendsa-server.cs.vt.edu/ODSA/Books/scgssm/csc230/spring-2026/CS_Courses/html/BinarySearch.html**